

Hongrui (Sam) Sheng

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EDUCATION

Fudan University, Shanghai, China

Expected 06/2027

Bachelor of Science in Chemistry

- GPA: 3.84/4.00 | Rank: 10/95 | GRE: 327 (V 157, Q 170, AW 4.0)
- **Key Coursework:** Quantum Chemistry (H): A+; Organic Chemistry II (H): A; Physical Chemistry II (H): A

Stanford University, California, USA

06/2025 - 08/2025

International Honors Program Participant

- STATS117 Introduction to Probability Theory: A+; CS106B Programming Abstractions: A

PUBLICATION

1. P. Walther[†], **Hongrui Sheng[†]**, X. Liu[†], B. Feng, R. Coyle, X. Yan, K. Smith, H. Kayal, S. C. Pal, and Z. Zheng*, “Predicting Scale-Up of Metal-Organic Framework Syntheses with Large Language Models,” *arXiv preprint*, 2026.
2. C. Lei, Y. Meng, **Hongrui Sheng**, S. Wang*, X. Wang, and M. Gong*, “The Interplay of Ions and Surfaces during Deposition/Dissolution Equilibrium Enables Efficient Neutral-pH Biomass Electrooxidation,” *Unpublished manuscript*, 2026, Submitted.

RESEARCH EXPERIENCE

Remote Research Assistant | Professor Zhiling (Zach) Zheng

Since 04/2026

Predictive Synthesis of Framework Materials

- Preparing figures and supporting information for “Predicting Scale-Up of Metal-Organic Framework Syntheses”.

Independent Research Program Leader, Wangdao Program | Professor Ming Gong

Since 05/2025

AI-aided tailoring of electrocatalysts for the selective valorization of glycerol

- Curated literature datasets and engineered novel descriptors using β function-inspired biasing factors to encode complex chemical interactions for downstream model training.
- Benchmarked machine learning models (GBR, RF, etc.) integrated with heuristic optimization (GA/PSO) for high-throughput screening of optimal metal alloy combinations.
- Assessed thermodynamic stability of candidate structures via ASE/MACE, with predictions validated against experimental wet-lab results.

Trainee researcher | Professor Ming Gong

03/2024 - 04/2025

Biomass Valorization via Electrocatalysis

- Executed multi-step protocols for catalyst fabrication, post-synthesis bleaching, and electrochemical testing.

SELECTED PROJECTS

Automated Computational Workflow for S_N2 Mechanisms | [Codes on GitHub](#)

11/2025 - 12/2025

Computational Chemistry Course Project, AI-assisted

- Built a **Python** pipeline automating **Gaussian** workflows for amine methylation, and applied **SISSO-inspired** regression ($R^2 > 0.95$) linking electronic (Q_N) and topological (κ_3) descriptors to activation energies.

AWARDS

Outstanding Student of Fudan University

2025

Second Prize, Syensqo Scholarship, Syensqo (formerly Solvay)

2025

Second Prize, Scholarship for Outstanding Students, Fudan University

2024, 2025

SKILLS & ADDITIONAL INFORMATION

Programming & Tools

Python, L^AT_EX, C++; Computational Chemistry (Psi4, Gaussian, ASE, MACE)

Languages

Chinese (Native), English (TOEFL iBT 107/120), Japanese (JLPT N1)

Leadership

Founding/Executive President, Fudan Integrative Medicine Assoc. & TCM Club (2024)

Interests

Hobby Aquarist, Rock and Classical Music, Traditional Chinese Medicine